

Rahul Yedida

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hello@ryedida.me

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EDUCATION

North Carolina State University

Ph.D. Computer Science

Raleigh, NC

Aug 2019 - Jul 2024

- Advisor: *Dr. Tim Menzies*
- Dissertation: [Guidelines for the Application of Neural Technologies in Software Analytics \(or: How to Do More with Less in SE\)](#)

PES University

B.E. Computer Science – Advisor: Dr. Snehanshu Saha

Bangalore, India

Aug 2015 - May 2019

EMPLOYMENT

LexisNexis Legal & Professional

Senior Data Scientist I

Raleigh, NC

May 2024 - Present

- Improved customer-facing product runtime by 24.8% and reduced peak memory usage duration by 21.3%. Helped improve complaint drafting results by 115% and motion drafting by 28.4% of usefulness.
- Tech lead for migration effort from RAG to agentic drafting system and subsequent maintenance.
- Developed fast passage filtering approach based on contrastive loss.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University

PhD Student

Raleigh, NC

Aug 2019 - Jul 2024

- **State-of-the-art hyper-parameter optimization:** Proposed a novel hyper-parameter optimization method that outperforms prior SOTA and is 36.4% faster.
- **Better, faster deep learning for SE:** Improved defect prediction by up to 123% (F-1 score), code smell detection by up to 30% (AUC), issue lifetime prediction by up to 76% (accuracy), automated microservice partitioning by up to 285% (modularity) compared to prior SOTA.
- **Semi-supervised learning:** Achieved state-of-the-art results (up to 100% improvement in AUC) on static code warnings analysis using 10% of the labels.
- **Teaching:** Teaching assistant for 830 students in total, over 5 semesters, for CSC 230 (C and Software Tools), CSC 510 (Software Engineering), and CSC 591/791 (Automated Software Engineering)

Amazon

Software Dev Engineer Intern

New York, NY / Bellevue, WA

May 2023 - Aug 2023

- Implemented profile locks for Prime Video on Echo Show devices.
- **Technology:** React Native, TypeScript

Software Dev Engineer Intern

May 2022 - Jul 2022

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners (DSPs).
- **Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

RECENT PUBLICATIONS

See full list on [Google Scholar](#).

1. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering*, 51(6).
2. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM*, 66(2), 75-81.
3. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering*, 49(4), 2856-2872.
4. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications*, 219, 119673.
5. **Yedida, R.**, Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR)*.

6. **Yedida, R.**, & Menzies, T. (2021). On the Value of Oversampling for Deep Learning in Software Defect Prediction. *IEEE Transactions on Software Engineering*, 48(8), 3103-3116.
7. Agrawal, A., Yang, X., Agrawal, R., **Yedida, R.**, Shen, X., & Menzies, T. (2021). Simpler Hyperparameter Optimization for Software Analytics: Why, How, When?. *IEEE Transactions on Software Engineering*, 48(8), 2939-2954.
8. Yang, X., Chen, J., **Yedida, R.**, Yu, Z., & Menzies, T. (2021). Learning to recognize actionable static code warnings (is intrinsically easy). *Empirical Software Engineering*, 26(3), 1-24.
9. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2021). Lessons learned from hyper-parameter tuning for microservice candidate identification. *Proceedings of the thirty-sixth IEEE/ACM International Conference on Automated Software Engineering (ASE)*.
10. **Yedida, R.**, & Saha, S. (2021). Beginning with Machine Learning: A Comprehensive Primer. *The European Physical Journal Special Topics*, 230(10), 2363-2444.
11. Saha, S., Nagaraj, N., Mathur, A., **Yedida, R.**, & Sneha, H. R. (2020). Evolution of novel activation functions in neural network training for astronomy data: habitability classification of exoplanets. *The European Physical Journal Special Topics*, 229(16), 2629-2738.
12. **Yedida, R.**, Saha, S., & Prashanth, T. (2020). LipschitzLR: Using theoretically computed adaptive learning rates for fast convergence. *Applied Intelligence*, 1-19.
13. Sridhar, S., Saha, S., Shaikh, A., **Yedida, R.**, & Saha, S. (2020, July). Parsimonious Computing: A Minority Training Regime for Effective Prediction in Large Microarray Expression Data Sets. In *2020 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8).
14. Khaidem, L., **Yedida, R.**, & Theophilus, A. J. (2019, November). Optimizing Inter-nationality of Journals: A Classical Gradient Approach Revisited via Swarm Intelligence. In *International Conference on Modeling, Machine Learning and Astronomy* (pp. 3-14). Springer, Singapore.

FUNDING

\$5,000, Google Cloud Academic Research Grant, Feb 2022

SERVICE TO PROFESSION

Contributing Member, Python Software Foundation (Mar 2021 - Present)

Guest Editor, IEEE Software SI: The Impact of AI on Productivity and Code 2025; ASEj SI: Replications and Negative Results (RENE) 2025; EMSE SI: Replications and Negative Results (RENE) 2025

Workshop Co-Chair, Intelligent Software Engineering (ASE '25, ICST '26); Replications and Negative Results (ASE '24)

PC Member, AAAI 2025-2027; FSE 2026-2027; ICSE 2026-2027; EDM 2026; ASE Industry Track 2026; AI Foundation Models and Software Engineering (FORGE) 2024; ICSME Artifact Evaluation Track, 2021-2023; ASE Artifact Evaluation Track, 2022; International Conference on Modeling, Machine Learning, and Astronomy (MMLA), 2019

Reviewer, TSE 2025-2026; NeurIPS 2023, 2025; EMSE 2021, 2025; ASEj 2025; ICML 2024-2025; ICLR 2024-2025; NCAA 2023-2025; TMLR 2024; Neural Processing Letters 2023-2024; Artificial Intelligence Review 2023; J. Big Data, 2023; ASE 2023; IEEE SSCI 2020

HONORS

Distinguished Reviewer, ICSE 2026

Mar 2026

Google Developer Expert - ML and GCP

Mar 2025 - Present

The Google Developer Experts program is a global network of highly experienced technology experts, influencers, and thought leaders who have expertise in Google technologies, are active leaders in the space, natural mentors, and contribute to the wider developer and startup ecosystem.

Google Cloud Champion Innovator - Cloud AI/ML

Oct 2022 - Mar 2025

Champion Innovators are a global network of more than 500 professionals, nominated by Googlers, who are technical experts in Google Cloud products and services. Each Champion specializes in one of nine different technical categories. In 2025, the Champion Innovators program was merged into the Google Developer Experts program.

Google Cloud Research Innovator

Feb 2022

Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.

INVITED TALKS

Feb 2024, “*Improving deep learning performance using theoretical ML*” at BITS Pilani, KK Birla Goa Campus, India

RELEVANT PROJECTS

Q&A System for Zotero

Rust, SQLite, LanceDB

Built an academic question-answering and semantic search system based on the user's Zotero library. Wrote a custom PDF parser (mean: 1,900 pages/s; 125 MB/s; n = 330) for academic papers and a RAG library with support for multiple providers and tool calling.

Oct 2024 - Present

[GitHub](#) :: [crates.io](#)

RAISE

Python, Keras

Sole developer for a PEP8-compliant, ML Python package used by our research lab and others for replicable results. Downloaded 65k times.

Aug 2020 - Jun 2025

[GitHub](#) :: [PyPI](#)

Programmable Resumes

Python

Developed a specification and implementation for modular, customizable resumes with support for two popular LaTeX templates. Wrote this resume using this tool.

Aug 2023 - Mar 2025

[GitHub](#)

pysh

C++, TMLanguage

Developed a superset of Python that allows running Shell code natively, with a VS Code syntax highlighting extension.

May 2021 - Dec 2024

[GitHub](#)

Threaded Discussions Website

MongoDB, Node.js, React

Companion website for video calls that allows for Reddit-style, threaded discussions.

Feb 2021 - Jun 2021

[GitHub](#)

Google/Meta Data Mining

Python, Keras

Data science project to use Google Takeout and Meta user data to suggest products to advertise to a user from Amazon best-sellers using DistilGPT-2, and achieved 0.6 F-1 score.

Feb 2021 - May 2021

[GitHub](#)

NearConnect

iOS, SwiftUI

iOS app to connect with people nearby using multicast peer-to-peer connections.

Nov 2020 - Mar 2021

[GitHub](#) :: [App Store](#)

Novel Drug Repurposing Hypotheses

Python, PyTorch

Identified novel drug repurposing hypotheses using text mining of radio transcripts, and verified results using a knowledge graph.

Oct 2019 - Feb 2020

[GitHub](#)

Personalized Chatbot

Python, Keras

Fine-tuned a GPT-2 345M model on 730k messages from Telegram logs to create a personalized chatbot.

May 2019 - May 2019

[GitHub](#)

Intelligent Tutoring System

Python

Implemented an Intelligent Tutoring System backend using Bayesian Knowledge Tracing and a novel question selection algorithm.

Sep 2018 - May 2019

[GitHub](#)

JournalBear

JavaScript, Electron

Cross-platform journal application with AES-256 encryption. Rated 4/5 by Softpedia.

Jan 2017 - Feb 2019

[GitHub](#) :: [Softpedia](#)

Astronomy Image Restoration

Python, Keras

Developed a machine learning approach to restore astronomical images affected by PSF anisotropy and smearing in crowded-field photometry, improving data retention and accuracy in differential imaging analysis of long-baseline optical time series.

Aug 2018 - Nov 2018

[GitHub](#)

Human Activity Data Project

Python, Keras

Collected personal activity data for 9 months, grouped tasks into 21 categories. Analyzed most productive hours of the day.

Oct 2018 - Nov 2018

[GitHub](#)

Results Scraper

MongoDB, Express.js, React, Node.js

Website for scraping university examination results and displaying charts and printable reports, with caching.

Mar 2018 - Aug 2018

[GitHub](#)

Video Sharing Website

Oct 2017 - Dec 2017

MySQL, Express.js, React, Node.js, Sass, Elasticsearch[GitHub](#)

Simplified implementation of a video-sharing website with subscriptions and custom searching.

Web development projects

Jan 2017 - Nov 2017

MongoDB, Express.js, React, Node.js, Sass, D3.js[CodePen](#)

Projects include URL shortener, rogue-like dungeon crawler game, voting application, Simon game, land surface temperature heatmap, and mapping meteorite impacts across the globe.

Xtreme Calculations

Apr 2013 - Oct 2017

VB.NET, Python[Softpedia](#)

Windows math software to solve scientific and mathematical problems, with over 30k downloads across multiple sites.

SKILLS**Languages:** Rust, Python, TypeScript, Java, C++, Gleam**ML/AI:** Keras, PyTorch, Hyper-parameter optimization**Frameworks:** React, Flask, Litestar, FastAPI**Databases:** LanceDB, MySQL, MongoDB, DynamoDB**Cloud:** Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)